Ex. no: 2
DATE: 06/08/26

STUDY OF NETWORK CABLES

AIM:

Study of different types of Network cables

a) Understand different types of network cable:

Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable.

Cable Type	Category	Maximum Data Transmissions	Advantages / Disadvantages	Application / Use
UTP	Category 3	10 bps	Advantages: * Cheaper in cost * Easy to install as they have a smaller overall diameter. Disadvantages: * More prone to (EMI) Electromagnetic interference and noise	10 Base -T Ethernet
	Category 5	Upto 100 Mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	1 Gbps		Fast Ethernet, Gigabit Ethernet
STP	Category 6, 6a	10 Gbps	Advantages: * Shielded * Faster than UTP * Less susceptible to noise and interference	Gigabit Ethernet, 10 G Ethernet, (55 m) widely used in data centres.

SSTP	Category 7	10 Gbps	Disadvantages: Expensive Greater installation effort	Gigabit Ethernet 10G Ethernet (600 m)
Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	High bandwidth Immune to interference Low Loss bandwidth versatile Disadvantages Limited distance Cost Size is bulky	Speed of signal is 500 m Television network High speed Internet connections
Fibre optics cable	Single mode Multi mode	100 Gbps	Advantages: High Speed High bandwidth High security Long distance	Maximum distance of fibre optics cable is around 100 meters.

b) Make Your Own Ethernet Cross - Over Cable / Straight cable

Tools and parts needed.

* Ethernet cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.
* A crimping tool. This is an all in one networking tool shaped to push down the pins in the plug & strip and cut the shielding off the cables.
* Two RJ45 plugs
* Optional two plug shields.

PROCEDURE :

Step 1 : To start construction of the device, begin by threading shields onto the cable.

Step 2 : Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3 : After you will need to untangle the wires, there should be four twisted pairs. Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.

Step 4 : Once the order is correct, bunch them together in a line, and if there any that stick out further than others. Snip them back to even level. The difficult aspect is placing those into the RJ45 plug without messing up the order. To do so hold the plug with the clip side facing away from you and have the golden pins facing towards you.

Step 5 : Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding and if it isn't, that means you stripped off too much shielding. Simply snip the cables back a little more.

Step 6 : After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

Step 7 : Lastly, repeat for the other end using diagram B / diagram A.

RESULT :

Thus, the different type of network cable has been studied and implemented successfully.